E-Commerce Sales & Customer Analytics

What five years of order data says about growth, discounts and delivery

Atharv Bajpai, Data Analyst | Power BI project | November 2026

Summary

I analysed 138,116 orders from 24,911 customers, covering January 2021 to December 2025, and built a four-page Power BI dashboard from them. This report goes through the dashboard page by page and ends with what I would do about the findings.

The business is profitable but it has stopped growing. Revenue is $177.1M and profit is $76.1M, a margin of 43%. Yearly revenue was $35.7M in 2021 and $35.1M in 2025, and orders sat at about 27.6K every year in between. Nothing is broken, but nothing is improving either.

Revenue	Profit	Profit margin	Orders	Customers	Avg order value
$177.1M	$76.1M	43.0%	138,116	24,911	$1,283

Five things stood out, and they shape the recommendations at the end:

- About one order in six never completes. 24,557 orders (17.8%) were cancelled, returned or left pending, which is $32.9M of booked revenue.
- Heavy discounts wipe out profit. Items sold at 40% off or more earn a 10.1% margin, compared with 50.7% for items discounted under 10%.
- Late deliveries cost ratings. Delayed orders average 3.22 stars against 3.73 for on-time ones, and about 15% of completed orders arrive late.
- November and December together bring in roughly a quarter of the year's revenue.
- The 32% of customers who order seven times or more generate 47% of revenue.

How the data was set up

The dashboard sits on four linked tables (customer_master, orders, order_items and product_catalog) plus a calendar table. Order-level numbers such as revenue, orders and delivery come from the orders table. Product, category and discount numbers come from order_items, and customer details come from customer_master so that each customer is counted once.

1. Business overview

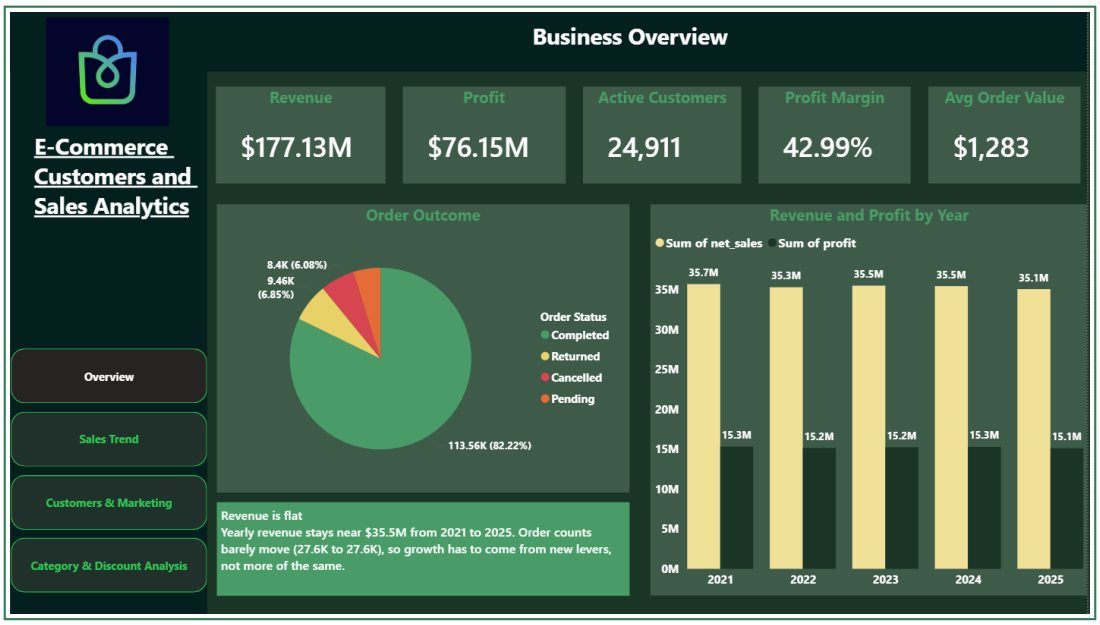


Figure 1. Overview page of the dashboard.

Out of every 100 orders, about 82 are completed. Another 7 are returned, 6 are cancelled and 5 are still pending. Together those 24,557 orders represent $32.9M of revenue and $16.4M of profit that the business never actually collects.

The revenue and profit bars barely change from year to year. Revenue stays between $35.1M and $35.7M and profit between $15.1M and $15.3M. Order counts are just as flat, 27.6K in 2021 and 27.6K in 2025, so the lack of growth is not a pricing effect. The business simply isn't selling more.

2. Sales trends

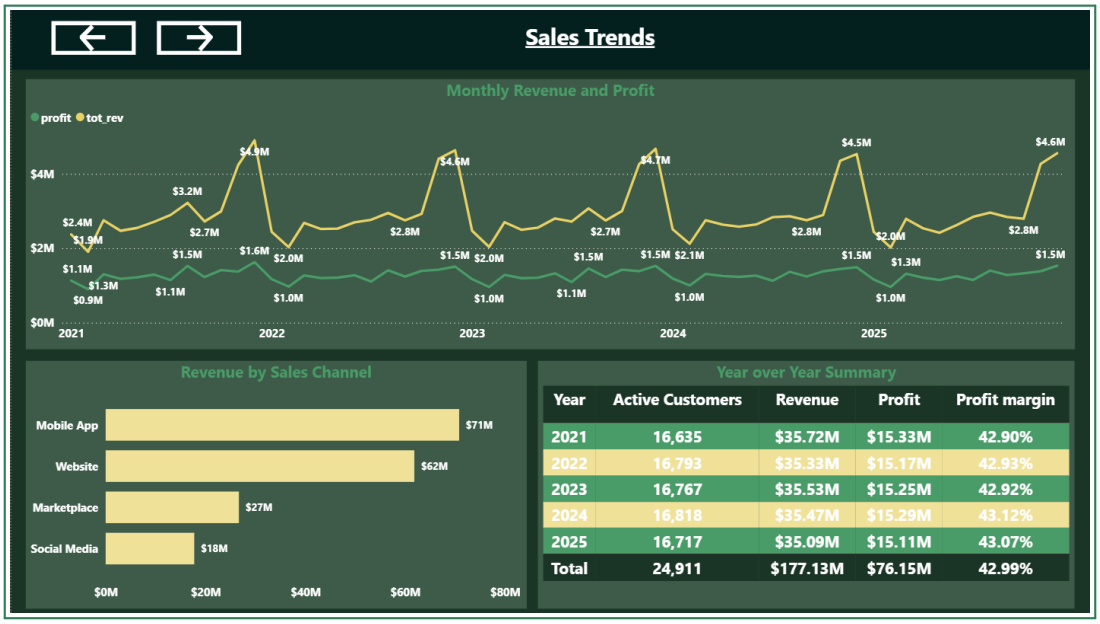


Figure 2. Sales Trends page: monthly revenue and profit, sales channels and the year-over-year summary.

The monthly line is the most interesting chart in the dashboard. For most of the year revenue stays between about $2.0M and $3.2M. Then every November and December it jumps to $4.5M to $4.9M, and in January it drops straight back. This happens in all five years. Averaged across the years, November brings in $4.3M and December $4.7M, against about $2.7M in a typical month, so those two months account for roughly 25% of annual revenue. Profit follows the same shape at a steady margin.

By channel, the Mobile App leads with about $71M (40% of revenue), then the Website with $62M, Marketplace with $27M and Social Media with $18M. Average order value is close to $1,280 on every channel, so a channel is bigger only because it gets more orders, not because customers spend more there.

The year-over-year table backs up the overall picture. Active customers hold at 16.6K to 16.8K a year and margin stays between 42.9% and 43.1%.

Because so much depends on the last two months of the year, stock levels, delivery capacity and payment systems need to be ready before November. The January drop is also a natural moment for win-back offers.

3. Customers and marketing

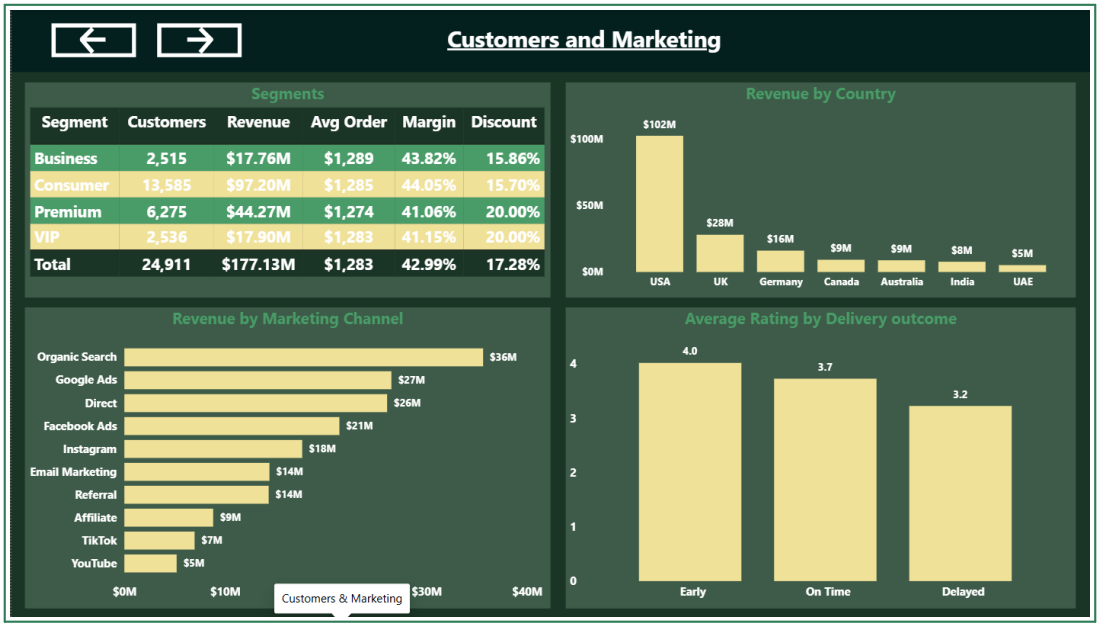


Figure 3. Customers and Marketing page: segments, country, marketing channel and rating by delivery outcome.

Segment	Customers	Revenue	Avg order	Margin	Avg discount
Business	2,515	$17.76M	$1,289	43.82%	15.86%
Consumer	13,585	$97.20M	$1,285	44.05%	15.70%
Premium	6,275	$44.27M	$1,274	41.06%	20.00%
VIP	2,536	$17.90M	$1,283	41.15%	20.00%
Total	**24,911**	**$177.13M**	**$1,283**	**42.99%**	**17.28%**

I expected Premium and VIP customers to be worth more per order, and they are not. Their average order is $1,274 to $1,283, the same as everyone else, yet they get a 20% average discount against about 15.7% for Consumer and Business customers. As a result their margin is roughly 3 points lower (41.1% against 44.1%). From this data, the extra discount is not buying larger baskets.

Geographically the business leans heavily on one market. The USA brings in $102M, about 58% of revenue, with the UK at $28M and Germany at $16M. Organic Search is the biggest marketing channel at $36M, ahead of Google Ads ($27M) and Direct ($26M), while TikTok ($7M) and YouTube ($5M) are the smallest. Customers who order four or more times produce 91% of revenue, and only 524 customers ordered once.

4. Category and discount analysis

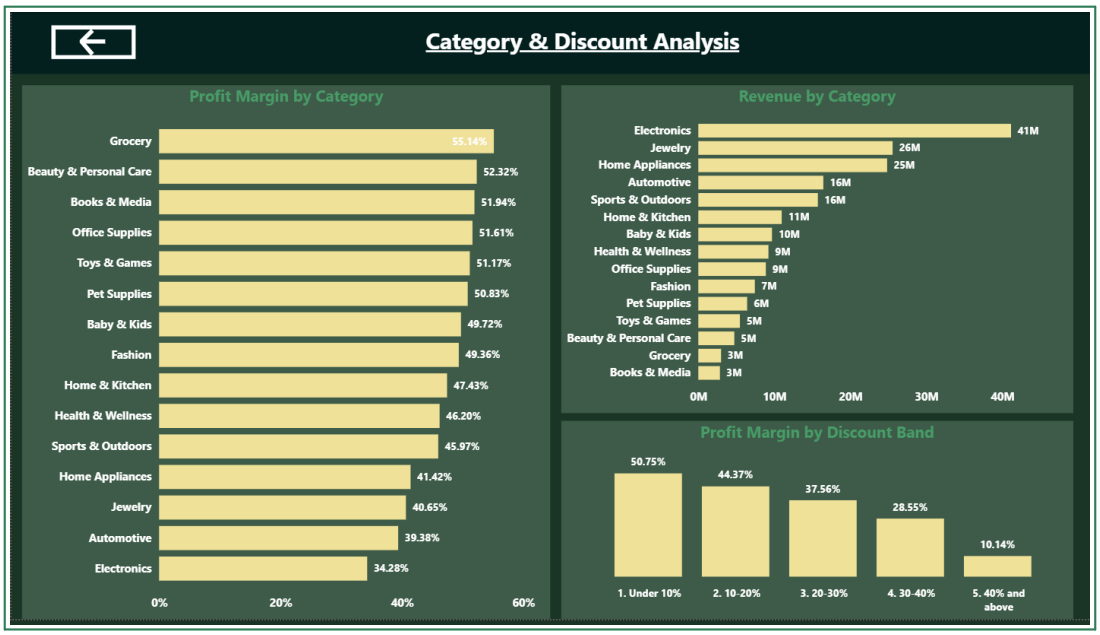


Figure 4. Category and Discount Analysis page.

The biggest categories are not the most profitable ones. Electronics brings in $41M, about 21% of item revenue, but has the lowest margin at 34.28%. Jewelry ($26M, 40.65%), Home Appliances ($25M, 41.42%) and Automotive ($16M, 39.38%) are also below 42%. At the other end, Grocery (55.14%), Beauty & Personal Care (52.32%) and Books & Media (51.94%) have the best margins but are small. Among brands, Samsung, LG and OnePlus earn the most profit, though each has a margin near 35%.

The discount chart shows the clearest pattern in the whole analysis. Margin falls at every step as discounts get deeper:

Discount band	Margin
Under 10%	50.75%
10-20%	44.37%
20-30%	37.56%
30-40%	28.55%
40% and above	10.14%

Items discounted 30% or more bring in $25.9M of revenue but only $5.2M of profit. As a rough illustration, if those items earned the margin of the 20-30% band instead, profit would be about $4.6M higher. That is an estimate, not a forecast, because customers may not buy at all without the deeper discount.

5. Delivery, returns and satisfaction

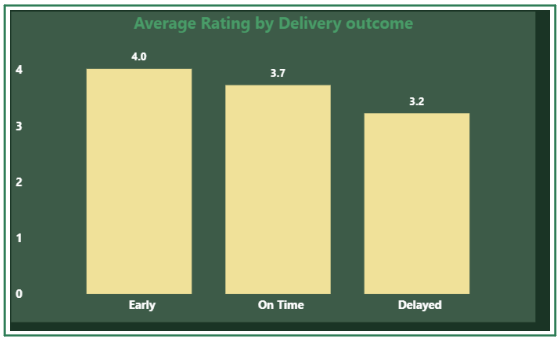


Figure 5. Average rating by delivery outcome (Customers and Marketing page).

Ratings follow delivery speed closely: 4.02 for early orders, 3.73 for on-time and 3.22 for delayed. About 14.9% of completed orders arrive late. The rate is almost the same for Same Day shipping (14.5%) and Economy (14.9%), which suggests the delays start in the warehouse or with the carrier and not in the shipping option the customer chose.

Returns tell a similar story. Of 9,462 returns, 5,868 (62%) were for wrong, damaged, defective, late or not-as-described items, all of which the company can control. Payment failures are steady at about 7.5% across every payment method.

What I would do first

1. Put a ceiling on discounts. Cap them at 30% and require approval for anything at 40% or more. This is the largest and clearest profit leak in the data.
2. Replace the blanket 20% discount for Premium and VIP customers with targeted offers. Right now it lowers margin without raising order size.
3. Prepare for November and December and plan January win-back offers. A quarter of annual revenue depends on those two months.
4. Look into delivery delays at the warehouse and carrier level, and set a target below 10% late. Delays hurt ratings and probably feed into returns.
5. Add packing and quality checks. Most returns come from mistakes the company makes itself.
6. Retry failed payments and send cart-recovery messages. 7.5% of payments fail and 4.8% of orders are left pending.
7. Review Electronics supplier costs or bundle accessories with it. It is the largest category and has the lowest margin.
8. Build a loyalty programme for customers with four or more orders, and test markets beyond the USA, since growth is not going to come from the current base alone.

Limits of this analysis

- Revenue is net sales as charged, so it includes tax and shipping. Profit is net sales minus product cost and shipping cost.
- Orders come in several currencies (USD, GBP, EUR and others). I added them up as given, treating them as dollars, because the dataset summary does the same. No conversion was applied.
- Product-level figures use order lines, whose total is about 8% higher than the order-header totals, and 92% of orders match exactly. The headline numbers use the orders table.
- There is no marketing spend in the data, so I could not measure return on investment by channel. The results are also very uniform across segments, channels and ages, so the dataset may be synthetic and the patterns are best read as a demonstration of method.